KubeCon CloudNativeCon North America 2026

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Breaking up Is Hard, but the Future Is with Helm 4

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Who we are

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Helm is a community project. This is our view from inside it.

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PART ONE

What are we breaking up with?



What Helm still is

The package manager for Kubernetes

CHART

A versioned package of Kubernetes resources

HELM

Installs and upgrades applications

RELEASE

A tracked installation in a cluster

Create · share · install · upgrade · roll back

helm.sh/docs/intro/introduction/

Developers rate Helm as mature

2019

Helm 3.0 released

94%

rated Helm maturity 4 or 5 stars

Maturity meant stability and reliability in the survey.

CNCF Q1 2026 Technology Radar · developers familiar with Helm

The cost of that compatibility promise

Public SDK APIs

Architectural limits

Accumulated debt

A major version makes room to change the interfaces that held new capabilities back.

Most of your investment stays

CHARTS

Chart API v2 remains supported.

RELEASES

Helm 4 can manage Helm 3 releases.

Start with a new binary and a representative test environment.

The Helm 3 support clock

Nov 2025

Helm 4 released

9 Sep 2026

Final limited Helm 3 minor

10 Feb 2027

Helm 3 security fixes end



Use the remaining support window to test and roll out.

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PART TWO

What changed in Helm 4?



Four foundations for the next chapter

01 Extend

Typed plugins and an optional Wasm runtime

02 Operate

Apply semantics and status watching

03 Distribute

OCI digests and repeatable packages

04 Evolve

A versioned SDK and the Charts v3 path

Plugins have a clearer shape

01

CLI

Add a Helm subcommand

02

Getter

Fetch charts from additional locations

03

Post-renderer

Transform rendered manifests

Optional Wasm runtime. Existing plugins continue to work.

A few integrations do change

HELM 3	HELM 4
POST-RENDERER	
`--post-renderer ./tool`	`--post-renderer my-plugin`
REGISTRY LOGIN TARGET	
`https://registry.example`	`registry.example`

Search CI scripts for both changes.

Apply behavior remembers the release

New Helm 4 release

Server-side apply by default

Existing Helm 3 release

Keeps client-side apply on upgrade / rollback

Override explicitly with --server-side when you choose.

Better signals while you wait

1

Deploy

Create or update

2

Observe

kstatus watches

3

Decide

Report status

Try complex, multi-resource rollouts and tell us what is confusing.

Pin the chart you meant to install

```
oci://registry.example.com/charts/app
@sha256:abc123…
```

Digest-based installs verify the exact OCI artifact.

Content-based caching and reproducible chart archives improve repeatability.

Small changes you will notice

Multi-document values

Split values across YAML documents.

Clearer errors

More actionable failures.

`--atomic`	→	**`--rollback-on-failure`**
`--force`	→	**`--force-replace`**

A foundation for the next chart format

Charts v2	Charts v3
Supported today	**Experimental**

Versioned SDK APIs make this evolution possible.

This is a feedback opportunity, not an upgrade requirement.

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PART THREE

How do we move forward?



Upgrade in five deliberate steps

01 **Test representative charts and releases**

02 **Exercise your CI / CD commands**

03 **Check plugins and post-renderers**

04 **Verify registry login and OCI installs**

05 **Roll out the Helm 4 binary**

Make the first upgrade boring

INVENTORY	`helm list -A`
VALIDATE	`helm lint ./chart`
RENDER	`helm template canary ./chart`
CANARY	`helm upgrade --install canary ./chart -n test`
OBSERVE	`helm status canary -n test`

Then test an existing Helm 3 release, a failure, and a rollback.

There is a way in for you

01

Try

Test a Helm 4 workflow and share a precise reproduction.

02

Talk

Join #helm-dev or the public weekly meeting.

03

Build

Open a docs change, issue, plugin, or project PR.

helm.sh/community

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TAKEAWAYS + Q&A

The future is with Helm 4

Questions?

Keep the charts. Test the workflow. Join the project.



Sources and further reading

Helm 4 overview

helm.sh/docs/overview/

Plugin migration

helm.sh/docs/plugins/migrate/

Helm 3 end-of-life timeline

helm.sh/blog/helm-v3-end-of-life/

Helm community

helm.sh/community/